

Plan-Ahead Resource Allocation Model

Mixed-Integer Second-Order Cone Program (MISOCP)
Multi-Tenant Cluster Optimization

Capstone Project — Cluster Optimization Models

Contents

1 Overview	1
2 Sets	2
3 Parameters	2
4 Decision Variables	3
5 Constraints	4
5.1 Machine Activation	4
5.2 Exclusive Tenant Constraints	4
5.3 Shared Tenant Constraints	4
5.4 Mix Bonus Constraints	5
6 Objective	5
7 Uncertainty Model (Cantelli Bound)	6
8 Feedback Integration	6
9 Connection to Real-Time Model	6

1 Overview

The plan-ahead model is a **Mixed-Integer Second-Order Cone Program (MISOCP)** solved once per planning horizon. It answers:

Which machines should each tenant be assigned to for each planning period, given their estimated resource demand, machine pool, and feedback from the previous horizon?

What the Model Decides

- (1) Which additional machines (beyond the always-available pool) to activate.
- (2) Which machines to assign *exclusively* to exclusive tenants *per period* (adapts each period via feedback).
- (3) Which machines to assign to *shared* tenants per period (can change each period).

- (4) How to mix high-usage and low-usage shared tenants on the same machines.

Output

An ordered list of planning periods. Each period has a list of **tenant groups**. Each group specifies the tenant IDs and their assigned machine IDs for that period. All tenants appear in every period. The Cluster Manager calls the Real-Time model once per group per interval (multiple times per period), passing only the pre-filtered jobs and machines.

Two Solver Modes

Mode	<code>use_socp</code>	Capacity constraint
MILP	False	Hard linear bound
MISOCP	True	Cantelli probabilistic bound (distribution-free)

2 Sets

Symbol	Description
$\mathcal{T}$	All tenants: $\mathcal{T} = \mathcal{T}_e \cup \mathcal{T}_s$
$\mathcal{T}_e$	Exclusive tenants. Do not share machines with others.
$\mathcal{T}_s$	Shared tenants: $\mathcal{T}_s = \mathcal{T} \setminus \mathcal{T}_e$. May share machines.
$\mathcal{T}_s^h$	Heavy-demand shared tenants (avg demand $\geq$ median over $\mathcal{T}_s$)
$\mathcal{T}_s^l$	Light-demand shared tenants: $\mathcal{T}_s \setminus \mathcal{T}_s^h$
$\mathcal{M}$	All machines: $\mathcal{M} = \mathcal{M}_a \cup \mathcal{M}_b$
$\mathcal{M}_a$	Always-available machines ($ \mathcal{M}_a = A$). Always active, no activation cost.
$\mathcal{M}_b$	Additional machines ($\mathcal{M} \setminus \mathcal{M}_a$). Model decides activation.
$\mathcal{H}$	Planning periods (slots): $\mathcal{H} = \{0, 1, \dots, \mathcal{H} - 1\}$. Each period spans <code>period_steps</code> intervals.

3 Parameters

Machine Parameters

Symbol	Description
C_n	Resource capacity of machine n (abstract units)
π_n	Activation cost for additional machine $n \in \mathcal{M}_b$
C_n^{eff}	Effective capacity after feedback: $C_n^{\text{eff}} = C_n(1 - \alpha \bar{v}_n)$

Tenant Demand Parameters

Symbol	Description
u_{ih}	Expected demand of tenant i in period h
u_i^{max}	Peak demand of exclusive tenant i : $\max_{h \in \mathcal{H}} u_{ih}$
u_{ih}^{fb}	Feedback-adjusted demand: $u_{ih}^{\text{fb}} = u_{ih}(1 + \beta \bar{W}_i / \bar{W}_{\text{ref}})$
σ_{ih}^2	Demand variance: $\sigma_{ih}^2 = (\sigma_{\text{frac}} \cdot u_{ih})^2$

Uncertainty and Safety Parameters

Symbol	Description
ε	Tail probability for Cantelli bound (default 0.10 $\Rightarrow$ 90% safety)
κ	Cantelli factor: $\kappa = \sqrt{(1 - \varepsilon)/\varepsilon}$. For $\varepsilon = 0.10$: $\kappa = 3.0$
σ_{frac}	Demand uncertainty fraction (default 0.20)

Feedback Parameters

Symbol	Description
$\bar{v}_n$	Rolling SLA violation rate on machine n (from real-time layer)
$\bar{W}_i$	Average scheduling wait time of tenant i (from real-time layer)
α	Capacity shrinkage coefficient (default 0.5)
β	Demand inflation coefficient (default 0.3)

Objective Weights

Symbol	Description
λ_0	Infrastructure cost weight
λ_1	Fairness weight (maximize minimum satisfaction ratio σ)
λ_2	Mix bonus weight (reward heavy+light co-location)

4 Decision Variables

Variable	Domain	Description
e_{in}	$\{0, 1\}$	1 iff exclusive tenant $i \in \mathcal{T}_e$ is assigned to machine n (entire horizon)
z_n^{on}	$\{0, 1\}$	1 iff additional machine $n \in \mathcal{M}_b$ is activated
z_{nh}	$\{0, 1\}$	1 iff machine n is active in period h
y_{inh}	$\{0, 1\}$	1 iff shared tenant $i \in \mathcal{T}_s$ is assigned to machine n in period h
f_{inh}	≥ 0	Capacity of machine n allocated to shared tenant i in period h
σ	$[0, 1]$	Min demand-satisfaction ratio across all shared tenants (fairness measure)
t_{nh}	≥ 0	Cantelli slack — upper bound on weighted demand std. dev. (MISOCP only)
hH_{nh}	$\{0, 1\}$	1 iff machine n has ≥ 1 heavy-demand tenant in period h
hL_{nh}	$\{0, 1\}$	1 iff machine n has ≥ 1 light-demand tenant in period h
mix_{nh}	$\{0, 1\}$	1 iff machine n has both heavy and light tenants in period h

5 Constraints

5.1 Machine Activation

C_{aa} — Always Available. Always-available machines are always active:

$$z_{nh} = 1 \quad \forall n \in \mathcal{M}_a, h \in \mathcal{H} \quad (\text{C}_{aa})$$

C_{act} — Additional Machine Activation. Additional machines are active only if switched on:

$$z_{nh} \leq z_n^{\text{on}} \quad \forall n \in \mathcal{M}_b, h \in \mathcal{H} \quad (\text{C}_{act})$$

C_{zact} — Machine Active if Assigned. A machine must be active if any shared tenant is assigned to it:

$$z_{nh} \geq y_{inh} \quad \forall i \in \mathcal{T}_s, n \in \mathcal{M}, h \in \mathcal{H} \quad (\text{C}_{zact})$$

5.2 Exclusive Tenant Constraints

C_{excl1} — At Most One Exclusive Per Machine.

$$\sum_{i \in \mathcal{T}_e} e_{in} \leq 1 \quad \forall n \in \mathcal{M} \quad (\text{C}_{excl1})$$

C_{excl2} — Each Exclusive Must Be Assigned.

$$\sum_{n \in \mathcal{M}} e_{in} \geq 1 \quad \forall i \in \mathcal{T}_e \quad (\text{C}_{excl2})$$

C_{excl.cap} — Exclusive Capacity Coverage. Machines assigned exclusively to tenant i must together cover its peak demand:

$$\sum_{n \in \mathcal{M}} e_{in} \cdot C_n \geq u_i^{\text{max}} \quad \forall i \in \mathcal{T}_e \quad (\text{C}_{excl.cap})$$

C_{sep} — Exclusive-Shared Separation. A machine assigned exclusively to any exclusive tenant cannot be used by shared tenants:

$$\sum_{i \in \mathcal{T}_e} e_{in} + y_{jnh} \leq 1 \quad \forall j \in \mathcal{T}_s, n \in \mathcal{M}, h \in \mathcal{H} \quad (\text{C}_{sep})$$

5.3 Shared Tenant Constraints

C_{share} — Every Shared Tenant Assigned Minimum Machines Every Period. Each shared tenant must receive at least $m_{\min}$ machines per period (default $m_{\min} = 2$; configurable via `min_machines_per_tenant`):

$$\sum_{n \in \mathcal{M}} y_{inh} \geq m_{\min} \quad \forall i \in \mathcal{T}_s, h \in \mathcal{H} \quad (\text{C}_{share})$$

Setting $m_{\min} \geq 2$ guarantees resilience: tenants stay scheduled even when one machine fails or spikes.

C_{1a} — Capacity with Cantelli Safety Buffer.

$$\sum_{i \in \mathcal{T}_s} f_{inh} + \kappa \cdot t_{nh} \leq C_n^{\text{eff}} \cdot z_{nh} \quad \forall n \in \mathcal{M}, h \in \mathcal{H} \quad (\text{C}_{1a})$$

In MILP mode: same constraint without the $\kappa \cdot t_{nh}$ term (and t_{nh} is absent).

C_{1b} — Cantelli Cone (MISOCP only).

$$\sum_{i \in \mathcal{T}_s} \sigma_{ih}^2 \cdot y_{inh} \leq t_{nh}^2 \quad \forall n \in \mathcal{M}, h \in \mathcal{H} \quad (\text{C}_{1b})$$

This is a second-order cone (SOC) constraint. Together, C_{1a} + C_{1b} guarantee:

$$P[\text{total demand on machine } n \text{ in period } h \leq C_n^{\text{eff}}] \geq 1 - \varepsilon.$$

C₂ — Priority Link. A shared tenant can receive allocation only on machines it is assigned to:

$$f_{inh} \leq C_n \cdot y_{inh} \quad \forall i \in \mathcal{T}_s, n \in \mathcal{M}, h \in \mathcal{H} \quad (\text{C}_2)$$

C₃ — Demand Satisfaction. Every shared tenant must receive at least its feedback-adjusted demand every period:

$$\sum_{n \in \mathcal{M}} f_{inh} \geq u_{ih}^{\text{fb}} \quad \forall i \in \mathcal{T}_s, h \in \mathcal{H} \quad (\text{C}_3)$$

C₄ — Fairness. σ is the minimum demand-satisfaction fraction across all shared tenants:

$$\sigma \leq \frac{\sum_{n \in \mathcal{M}, h \in \mathcal{H}} f_{inh}}{\sum_{h \in \mathcal{H}} u_{ih}^{\text{fb}}} \quad \forall i \in \mathcal{T}_s \quad (\text{C}_4)$$

5.4 Mix Bonus Constraints

$\text{mix}_{nh} = 1$ iff machine n has at least one heavy *and* one light tenant in period h . This is a linearized AND gate:

$$\text{hH}_{nh} \geq y_{inh} \quad \forall i \in \mathcal{T}_s^{\text{h}}, n \in \mathcal{M}, h \in \mathcal{H} \quad (\text{Mix}_H)$$

$$\text{hL}_{nh} \geq y_{jnh} \quad \forall j \in \mathcal{T}_s^{\text{l}}, n \in \mathcal{M}, h \in \mathcal{H} \quad (\text{Mix}_L)$$

$$\text{mix}_{nh} \leq \text{hH}_{nh} \quad \forall n \in \mathcal{M}, h \in \mathcal{H} \quad (\text{Mix}_1)$$

$$\text{mix}_{nh} \leq \text{hL}_{nh} \quad \forall n \in \mathcal{M}, h \in \mathcal{H} \quad (\text{Mix}_2)$$

$$\text{mix}_{nh} \geq \text{hH}_{nh} + \text{hL}_{nh} - 1 \quad \forall n \in \mathcal{M}, h \in \mathcal{H} \quad (\text{Mix}_3)$$

6 Objective

$$\boxed{\min \lambda_0 \underbrace{\sum_{n \in \mathcal{M}_b} \pi_n z_n^{\text{on}}}_{\text{InfraCost}} - \lambda_1 \sigma - \lambda_2 \underbrace{\sum_{n \in \mathcal{M}, h \in \mathcal{H}} \text{mix}_{nh}}_{\text{MixTotal}}} \quad (1)$$

(1) **InfraCost** (minimize): activate as few additional machines as possible.

(2) **Fairness** σ (maximize): every shared tenant receives its fair share.

(3) **MixTotal** (maximize): reward co-locating heavy-demand and light-demand tenants.

7 Uncertainty Model (Cantelli Bound)

Why Cantelli?

We know only mean μ_{ih} and variance σ_{ih}^2 of tenant demand — not the full distribution. Cantelli’s inequality is *distribution-free*: it bounds the one-sided tail using only mean and variance.

Applied to the Capacity Constraint

Let $D_{nh} = \sum_{i \in \mathcal{T}_s} D_{inh}$ be the total random demand on machine n in period h , where D_{inh} has mean f_{inh} and variance $\sigma_{ih}^2 y_{inh}$.

By Cantelli’s inequality:

$$P[D_{nh} > \mu_{nh} + \kappa \sigma_{nh}^{\text{tot}}] \leq \frac{1}{1 + \kappa^2} = \varepsilon,$$

where $\mu_{nh} = \sum_i f_{inh}$ and $(\sigma_{nh}^{\text{tot}})^2 = \sum_i \sigma_{ih}^2 y_{inh}$.

Constraints $C_{1a} + C_{1b}$ together enforce $\mu_{nh} + \kappa \sigma_{nh}^{\text{tot}} \leq C_n^{\text{eff}} z_{nh}$, giving:

$$P[D_{nh} \leq C_n^{\text{eff}} z_{nh}] \geq 1 - \varepsilon.$$

For $\varepsilon = 0.10$, $\kappa = 3.0$: machines overflow at most 10% of the time under the worst-case demand distribution with the given mean and variance.

8 Feedback Integration

After the first planning horizon, the Real-Time layer reports $\bar{v}_n$ (SLA violation rate) and $\bar{W}_i$ (average wait time per tenant), which update the plan parameters:

$$C_n^{\text{eff}} = C_n(1 - \alpha \bar{v}_n) \quad (\text{shrink capacity of overloaded machines}) \quad (2)$$

$$u_{ih}^{\text{fb}} = u_{ih} \left(1 + \beta \min \left(2, \frac{\bar{W}_i}{\bar{W}_{\text{ref}}} \right) \right) \quad (\text{inflate demand for tenants with long waits, capped at } 2 \times) \quad (3)$$

The cap at 2 prevents extreme demand inflation from making the MISOCP infeasible. Default: $\alpha = 0.5$, $\beta = 0.3$, $\bar{W}_{\text{ref}} = 10$ intervals (calibrated to the simulated time-step, not wall-clock seconds).

9 Connection to Real-Time Model

The Cluster Manager uses the plan output as follows each simulation interval:

1. Resolve $h = (\text{sim_interval} // \text{period_steps}) \bmod |\mathcal{H}|$.
2. Get the group list for period h .
3. For each group g :
 - (a) Filter job queue: jobs with $\text{tenant_id} \in g.\text{tenant_ids}$.
 - (b) Filter nodes: machines with $\text{node_id} \in g.\text{machine_ids}$.
 - (c) Call `solve(group_jobs, group_nodes,  $W_t$ ,  $K$ , time_limit_ms)`.

(d) Place returned assignments; unplaced jobs return to queue with a wait-time bump.

The Real-Time model has **no knowledge** of tenants, plan-ahead, or machine assignments. It receives only the pre-filtered jobs and nodes, and returns the best placement given capacity.